

Searching for Remote Work

Matching, Bargaining, and Sorting

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Introduction

How does the feasibility of remote work influence the wage premiums of remote workers?

1. **Empirical Motivation:** Establish key stylized facts about remote work and wages.
2. **Theoretical Model:** Develop a search and matching model to explain these facts.
3. **Model Results & Implications:** Discuss the model's predictions and how they align with the data.

Positioning Our Contribution

- **Teleworkability Measurement (Dingel & Neiman 2020; Mongey et al. 2021):** We build on this by constructing a new index to capture the nuance of remote work feasibility.
- **Remote Work & Compensation (Bloom, Cullen, et al.):** This literature documents wage premiums. We link these premiums to teleworkability within a structural model.
- **Amenity Provision & Hedonic Wages (Hwang, Mortensen, Reed 1998; Morchio & Moser 2024):** These papers model job amenities. We contribute by linking the remote work amenity directly to productivity, with rich heterogeneity.

Empirical Motivation

Who Works From Home?

A simple comparison shows remote workers are different.

Characteristic	Non-WFH	WFH
Share of Labor Force (2020-23)	85%	15%
Average Age	44.2	44.7
Average Hourly Wage	\$27.95	\$44.20
Share with College Degree	39%	66%
Share with Postgraduate Degree	15%	26%

Key takeaway: Remote workers are, on average, higher-educated and have significantly higher raw earnings, pointing towards strong selection effects.

Stylized Fact I: A “Teleworkability” Premium

Workers in occupations more suitable for remote work earn higher wages.

- Regressing log wages on our **Teleworkability Index** shows a strong positive correlation.
- This premium remains highly significant even after controlling for a rich set of fixed effects.
- This suggests that the skills and tasks that make a job “teleworkable” are also skills and tasks that the market values highly.

Stylized Fact II: A WFH Premium Within Occupations

Even within the same occupation, workers who work from home earn more.

- After controlling for detailed occupation fixed effects, WFH indicator is associated with an **8.8% wage premium**.
- This cannot be explained by sorting across occupations alone.
- **The Puzzle:** Why does this premium exist? Is it unobserved worker skill, firm productivity, or a feature of the matching process? Our model is designed to answer this.

The Model

Model Environment: The Key Ingredients

Agents & The Match

- **Workers** (differ in skill h) and **Firms** (differ in remote efficiency ψ) search for each other in the labor market.
- When they meet, they bargain over a contract: a **wage** (w) and a **remote share** (α).

The Bargain & Technology

- The contract is determined by **efficient Nash bargaining** to maximize and split a joint surplus.
- Output depends on the interaction between worker skill, firm technology, and the remote share:

$$Y(\alpha \mid h, \psi) = A(h) \cdot [(1 - \alpha) + \alpha \cdot g(h, \psi)]$$

The Worker's Perspective: Value Functions

Value of Unemployment, $U(h)$

$$U(h) = \underbrace{b(h)}_{\text{Today's Benefit}} + \beta \left[\underbrace{(1-p)U(h)}_{\text{No Match}} + \underbrace{p\mathbb{E}_\psi[W(h, \psi)]}_{\text{Finds a Job}} \right]$$

- The worker's future depends on the job-finding rate (p).
- They anticipate the value, $W(h, \psi)$, they will get from a future match.

Value of Employment, $W(h, \psi)$

$$W(h, \psi) = \underbrace{U(h)}_{\text{Outside Option}} + \underbrace{\xi S(h, \psi)}_{\text{Share of the Surplus}}$$

- The worker receives their reservation value, $U(h)$, plus a share ξ of the match surplus $S(h, \psi)$.

Total Value of a Match, $J(h, \psi)$

The total value of a match is the present value of all future joint payoffs.

$$J(h, \psi) = \underbrace{\text{Joint Flow Value}}_{\text{Today's "Pie"}} + \beta \left[\underbrace{(1 - \delta)J(h, \psi)}_{\text{Match Continues}} + \underbrace{\delta U(h)}_{\text{Match Ends}} \right]$$

- If the match ends (with probability δ), the firm gets 0 and the worker gets $U(h)$.

Bargaining & Wages

Defining the “Pie”: Efficient Bargaining

- We assume **efficient bargaining** and quasi-linear utility, so the wage is a pure transfer.
- The joint flow value is simply output minus the non-wage utility cost:

$$\text{Joint Flow Value} = Y(\alpha^* \mid h, \psi) - c(\alpha^*)$$

- The **Match Surplus** is the net gain from this cooperation:

$$S(h, \psi) \equiv J(h, \psi) - U(h)$$

The “Pie” to be Split: Match Surplus

The **match surplus** is the total value created by the match. It’s the central object of the bargain and is determined by a Bellman equation:

$$S(h, \psi) = \underbrace{s(h, \psi)}_{\text{Today's Net Gain}} + \beta(1 - \delta)S(h, \psi) - \underbrace{\beta p \xi \mathbb{E}[S^+]}_{\text{Value of Worker's Outside Option}}$$

- The surplus today ($s(h, \psi)$) is the joint output minus the worker’s unemployment benefit.
- The future surplus is discounted, but is reduced by the value of the worker’s option to search for another job while unemployed.

Dividing the Pie: The Wage

The wage w^* is the tool used to divide the surplus. It is set to ensure the worker receives their bargained value. This involves two key principles:

1. **The Bargaining Outcome:** The worker's value from the job must equal their outside option (unemployment) plus their share (ξ) of the surplus.

$$W(h, \psi) = U(h) + \xi S(h, \psi)$$

2. **The Value of the Contract:** The worker's value is also the present value of the utility stream from the contract.

$$W(h, \psi) = \frac{u(w, \alpha) + \beta \delta U(h)}{1 - \beta(1 - \delta)}$$

The equilibrium wage w^* is the wage that makes these two equations hold simultaneously.

Model Results

How is the Remote Share (α^*) Determined?

The Efficient Bargain

The optimal remote share, α^* , is chosen to maximize the total match value. This boils down to a simple trade-off:

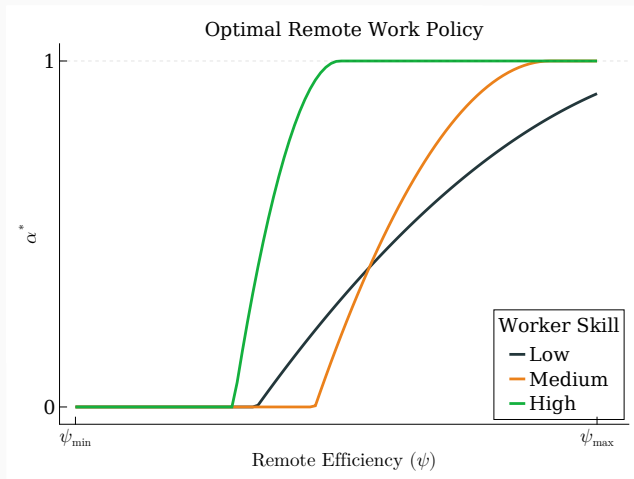
$$\underbrace{\frac{\partial Y}{\partial \alpha}}_{\text{Marginal Product of Remote Work}} = \underbrace{-\frac{\partial u}{\partial \alpha}}_{\text{Marginal Utility Cost of In-Office Work}}$$

The optimal remote share $\alpha^*(h, \psi)$ depends on the relative productivity of remote work, $g(h, \psi)$.

- If $g(h, \psi)$ is high $\implies$ Fully remote ($\alpha^* = 1$).
- If $g(h, \psi)$ is low $\implies$ Fully in-office ($\alpha^* = 0$).
- If $g(h, \psi)$ is intermediate $\implies$ Hybrid work ($0 < \alpha^* < 1$).

Visualizing the Optimal Remote Share

- This plot shows how the optimal remote share α changes with firm efficiency ψ skill h .
- The non-monotonic ordering and crossover highlight the complex interaction between baseline productivity and remote work efficiency



Conclusion

Explaining the Stylized Facts

- **Fact I (Teleworkability Premium):** The model can generate this if the skills (h) that are valued by the market ($A(h)$) are also those that make remote work productive ($g_h > 0$). The tasks that command high wages are the same ones that allow for remote work.
- **Fact II (Within-Occupation Premium):** The model provides two key channels for this:
 1. **Positive Assortative Matching:** Within an occupation, higher-skilled workers (unobserved in simple regressions) sort into firms with higher remote efficiency (ψ). This match is more productive, leading to both a higher wage and a higher remote share.
 2. **Compensating Differentials:** For a given worker, firms with lower remote efficiency (ψ) may need to offer different wages to attract them, creating a complex wage-amenity landscape.

What We Did

- Established two key stylized facts about the remote work wage premium.
- Developed a two-sided search model with bargaining over wages **and** remote work to explain these facts.
- Showed that **sorting and selection** are key mechanisms driving the observed premium.

Key Takeaway

The remote work wage premium is not a simple amenity-vs-wage tradeoff. It is an **equilibrium outcome** reflecting which workers match with which firms, and how their joint productivity is shaped by remote work technology.

Thank You!

Questions?

The value of being unemployed is the sum of unemployment benefits (b) and the expected value of finding a new job.

$$U(h) = b(h) + \beta \left[(1 - p)U(h) + p \int \max\{W(h, \psi), U(h)\} \frac{v(\psi)}{V} d\psi \right]$$

Which simplifies to:

$$U(h) = \frac{b(h) + \beta p \xi \mathbb{E}_{\psi}[S(h, \psi)^+]}{1 - \beta}$$

The equilibrium wage w^* is found by setting the two expressions for the worker's value $W(h, \psi)$ equal:

$$\frac{u(w^*, \alpha^*) + \beta\delta U(h)}{1 - \beta(1 - \delta)} = U(h) + \xi S(h, \psi)$$

Solving for the utility flow gives the target that the wage and amenity must deliver:

$$u(w^*, \alpha^*) = [U(h) + \xi S(h, \psi)] (1 - \beta(1 - \delta)) - \beta\delta U(h)$$

Appendix: Vacancy Creation & Market Tightness

Firm's Problem

Firms post vacancies until the marginal cost equals the expected marginal benefit.

$$\underbrace{c'(v(\psi))}_{\text{Marginal Cost}} = \underbrace{q}_{\text{Fill Prob.}} \cdot \underbrace{(1 - \xi) \mathbb{E}_h[S(h, \psi)^+]}_{\text{Expected Firm Surplus}}$$

Market Tightness, $\theta = V/L$

This condition determines the equilibrium level of market tightness, which in turn determines the job-finding (p) and job-filling (q) rates.

$$\theta = \left(\frac{1}{L} \int \left[\frac{(1 - \xi) \mathbb{E}_h[S(h, \psi)^+]}{\kappa_0} \right]^{1/\kappa_1} d\psi \right)^{\frac{\kappa_1}{\kappa_1 + \gamma}}$$

The Alternating Offers Game

- **Setup:** A worker and firm meet. In each period, one player is chosen to propose a contract (w, α) . Worker proposes with probability ξ .
- **Delay is Costly:** Future is discounted by $\beta(1 - \delta)$.
- **Equilibrium:** A player offers a contract that makes the other party just indifferent between accepting and waiting.

The Result

- This game leads to the **Nash Bargaining Solution**. The surplus is split according to the proposal probabilities:

$$\frac{\text{Worker's Share of Surplus}}{\text{Firm's Share of Surplus}} = \frac{\xi}{1 - \xi}$$

- This provides a theoretical foundation for the bargaining assumption in the main model.